

Rocket Elevators

ELEVATE SAFETY, SERVICE AND VALUE

INTERNAL DOCUMENT

Traction Elevator Troubleshooting Guide

Internal document. Last updated April 2026. Contact: Field Operations Manager, Rocket Elevators Ontario Division.

This guide is the primary troubleshooting reference for traction elevator systems in our fleet. It is organized by symptom: when a technician encounters a problem in the field, they look up the symptom here and follow the diagnosis and resolution steps. This document covers traction-specific issues only. For hydraulic elevator maintenance, see the Hydraulic Elevator Maintenance Procedures document.

Traction elevators account for the majority of our passenger elevator fleet, particularly in buildings over six stories. They also include our Freight Elevator-E (electric traction) variants and all Observation Elevators. ThyssenKrupp Elevator is our primary contractor for traction systems, and most escalation paths in this guide lead to them.

Important safety notice: Never bypass a safety device to test a theory. If a safety circuit is tripped, there is a reason. Our job is to find that reason, not to override it. Any technician who bypasses a safety device to restore service faces immediate review. TSSA takes safety circuit tampering seriously, and so do we.

Traction System Overview

A traction elevator works on a simple principle: an electric motor drives a **sheave** (a grooved pulley), and steel **hoist ropes** pass over the sheave connecting the car on one side to a **counterweight** on the other. The counterweight typically equals the weight of the car plus about 40% of the rated load, so gravity does most of the heavy lifting. The motor only needs to overcome the difference between the loaded car and the counterweight, which is why traction systems are far more energy-efficient than hydraulic systems for taller buildings.

Geared vs. Gearless

Our fleet includes both types:

- **Geared traction** systems use a gearbox between the motor and the sheave. These are common in older installations and buildings up to roughly 20 stories. The gearbox provides mechanical advantage, allowing a smaller motor, but introduces more moving parts and maintenance points. Many of our geared units are in buildings built before 2000.
- **Gearless traction** systems connect the motor directly to the sheave. The motor is larger and turns more slowly, but the elimination of the gearbox reduces maintenance, noise, and vibration. Most of our newer high-rise installations use gearless systems.

Machine Room vs. Machine-Room-Less (MRL)

Traditional traction elevators have a **machine room** at the top of the hoistway containing the motor, drive, controller, and governor. Technicians access these components in the machine room during diagnosis.

Machine-room-less (MRL) installations place the motor and drive in the hoistway itself, typically at the top, with the controller in a wall-mounted cabinet on the top landing. MRL systems are increasingly common in our newer installations because they save building space. However, they present unique troubleshooting challenges: access is more restricted, ventilation is different, and some diagnostic steps that are straightforward in a machine room require hoistway access in MRL configurations. When this guide references "the machine room," MRL technicians should understand this as the equivalent access point for their configuration.

Troubleshooting Methodology

Every troubleshooting call follows the same systematic approach. Skipping steps leads to misdiagnosis, wasted time, and repeat visits.

1. **Observe the symptom.** Talk to the building contact. What exactly is happening? When did it start? Is it intermittent or constant? Has anything changed recently (construction, power work, weather)? Write down the reported symptom in your own words before touching anything.
2. **Check the fault log.** Every modern controller maintains a fault log. Access it and record the last 10-20 fault entries, including timestamps. Patterns in fault codes are often more diagnostic than any single entry. Note: older relay-based controllers do not have digital logs; for these, check the position of safety relays and contactors.
3. **Isolate the system.** Traction elevator problems fall into four categories: mechanical (motor, sheave, ropes, brakes, guides), electrical (drive, controller, power supply, wiring), door system (operators, interlocks, sensors), and safety circuits (governor, buffers, safeties, switches). Determine which system the symptom points to before testing.
4. **Test systematically.** Start with the simplest possible cause and work outward. Check power supply before suspecting the drive. Check the door interlock before suspecting the controller. Use your multimeter, not your intuition.
5. **Document findings.** Record everything in the service report: symptom, fault codes, tests performed, measurements taken, root cause (confirmed or suspected), and action taken. Even if you cannot resolve the issue, thorough documentation saves the next person hours.
6. **Decide: field-fix or escalate.** Our field technicians handle specific categories of work (see "When to Escalate" at the end of this guide). If the problem falls outside our scope, document your findings and contact ThyssenKrupp to schedule the repair. A good diagnosis report from our technician means the contractor arrives prepared, which reduces downtime.

Motor and Drive Issues

The motor and drive are the heart of the traction system. The **drive** (also called the **variable frequency drive** or **VFD**) controls the speed and torque of the motor. Problems in this area affect car speed, smoothness, and whether the elevator runs at all.

Symptom: Motor Overheating

The motor has thermal protection that will shut the elevator down if temperature exceeds safe limits. If you arrive and the motor is hot to the touch or the thermal overload has tripped:

1. Check the **machine room ventilation**. Traction motors generate significant heat, and machine rooms require adequate airflow. Look for blocked vents, failed exhaust fans, or HVAC systems that have been turned off. In summer months, machine room temperatures in older buildings can exceed safe operating ranges. This is one of the most common causes of motor overheating in our fleet, and the fix is often as simple as ensuring the ventilation is working.
2. Check the **motor bearings** by listening for grinding, squealing, or rumbling sounds with the car running (if it can be safely operated). Bearing failure increases friction and generates heat. Note: do not attempt to replace or lubricate bearings; document and escalate.
3. Review recent **usage patterns**. Has the building experienced unusually heavy traffic (move-in/move-out, event)? Sustained high-duty operation can overheat motors that are correctly sized for normal use.
4. Check the **drive** for fault codes related to overcurrent or overload. A failing drive can push excessive current to the motor.
5. Inspect the **motor windings** visually for discoloration or burnt smell, which indicates insulation breakdown. This requires contractor repair.

Field action: Restore ventilation if that is the cause. For any other cause, take the car out of service and escalate to ThyssenKrupp. Do not reset the thermal overload and return the car to service without identifying the cause; the overload exists to prevent motor fires.

Symptom: Speed Irregularities

The car runs too fast, too slow, or "hunts" (oscillates around the target speed). Passengers may notice the car feels sluggish or surges.

1. Check the **encoder** (also called a **tachometer** or **pulse generator**). The drive uses the encoder to know how fast the motor is turning. If the encoder signal is noisy or dropping out, the drive cannot maintain correct speed. Look for loose connections at the encoder, contamination on the encoder wheel, or a damaged cable. Encoder problems are the most common cause of speed irregularities.
2. Check for **rope slip** on the sheave. If the ropes are slipping rather than gripping the sheave grooves, the car will not respond correctly to drive commands. Rope slip can indicate sheave groove wear, incorrect rope tension, or contamination (oil, grease) on the ropes or sheave. See "Rope and Sheave Problems" below.
3. Review the **drive parameters**. Drive faults or parameter drift (sometimes after a power event or firmware update) can cause speed control issues. This requires a technician with drive-specific training and is contractor work.
4. Check **power supply** quality. Voltage sags, swells, or phase imbalance can cause the drive to behave erratically. Use your power quality meter to verify voltage on all three phases.

Field action: Check and reseal encoder connections. Report findings and escalate drive or sheave issues to ThyssenKrupp.

Symptom: Excessive Vibration

Vibration felt in the machine room, transmitted through the ropes, or reported by passengers:

1. Check **motor mounting bolts** for looseness. Vibration can loosen mounting hardware over time, and a loose motor amplifies vibration.
2. Listen to the **bearings**. Worn bearings produce vibration that increases with speed.
3. Inspect the **sheave** for uneven groove wear. A worn sheave causes ropes to seat unevenly, creating vibration at the frequency of the sheave rotation.
4. For geared systems, check the **gearbox** for abnormal noise. Gear tooth wear or low lubricant levels cause vibration and noise. Check the oil level and look for metal particles in the oil, which indicate gear wear.
5. Check the **motor coupling** (geared systems). A worn or misaligned coupling transmits vibration between the motor and gearbox.

Field action: Tighten any loose mounting hardware. All other causes require contractor diagnosis and repair.

Symptom: Motor Won't Start

The elevator does not respond to hall calls. The car sits at a floor with doors closed and no movement:

1. Verify **power supply** to the motor and drive. Check all three phases at the disconnect. A blown fuse or tripped breaker on one phase will prevent the motor from starting (or cause it to attempt to start on two phases, which the drive should fault on).
2. Check the **main contactor**. The contactor connects power to the drive. If it is not pulling in, check the control circuit that energizes the contactor coil. A failed contactor is a common cause of "dead elevator" calls.
3. Check whether the **overload** has tripped. If so, investigate why before resetting (see Motor Overheating above).
4. Check the **drive** for fault codes. Many drive faults will prevent the motor from starting. Record all fault codes.
5. Verify the **safety circuit** is complete. An open safety circuit (any safety device in the chain is tripped) will prevent the elevator from running. Check the safety circuit status on the controller. Common safety devices that trip: governor switch, pit stop switch, car top stop switch, door interlocks, buffer switches.
6. For MRL systems, check whether the **emergency manual operation** switch has been left in manual mode after a previous service call. This is embarrassingly common and saves everyone time when checked early.

Field action: Reset breakers or overloads only after verifying the cause is resolved. Reseat loose connections. Report all findings. If the safety circuit is open, identify which device is tripped but do not bypass it. Escalate to ThyssenKrupp for drive faults, contactor replacement, or unresolved electrical issues.

Rope and Sheave Problems

All rope and sheave work is contractor work. Our field technicians perform visual inspections and report findings, but never adjust, replace, or tension ropes. This section helps technicians identify problems and provide accurate reports to ThyssenKrupp.

Hoist Rope Inspection

CSA B44 specifies the criteria for rope replacement. During visual inspections, check for:

- **Broken wires.** Count visible broken wires per **rope lay** (one complete twist of the strands around the core). CSA B44 specifies the maximum number of broken wires per lay before replacement is required. Even a few broken wires should be documented and reported. See the Safety Code Quick Reference for the specific numeric thresholds.
- **Diameter reduction.** Measure rope diameter with calipers and compare to the original diameter (recorded on the rope data tag or in the maintenance records). A reduction of more than a certain percentage from the nominal diameter indicates internal wear and requires replacement. Ropes wear from the inside as well as the outside.
- **Corrosion.** Surface rust is common and not always critical, but extensive corrosion weakens the wire strands. Look for pitting, flaking, or discoloration along the length of the rope. Corrosion is particularly common in hoistways exposed to moisture (buildings with rooftop machine rooms, coastal environments, or poor drainage).
- **Bird-caging.** This occurs when the outer strands bulge outward from the rope core, usually caused by a sudden release of tension. Bird-caging is a serious condition requiring immediate rope replacement.
- **Unequal tension.** All hoist ropes in a set should carry approximately equal load. Visually check that all ropes have similar sag when the car is at a mid-point in the hoistway. Significantly different sag indicates unequal tension, which accelerates wear on the tighter ropes and reduces traction. Rope tension equalization is contractor work.

Sheave Groove Wear

Over time, the hoist ropes wear grooves into the sheave. This is expected, but excessive or uneven wear creates problems:

- **Undercut grooves.** The bottom of the groove wears deeper than specified, reducing the contact area between the rope and the groove wall. This reduces traction and can cause rope slip.
- **Uneven groove wear.** If grooves wear at different rates, some ropes carry more load than others, accelerating the wear cycle.
- **Groove measurement.** Use a groove gauge to measure the groove profile against the manufacturer's specifications. Record measurements for each groove and report to ThyssenKrupp. They will determine whether the sheave can be re-grooved (machined to restore the correct profile) or needs replacement.

Compensation Rope/Chain Issues

Taller buildings use **compensation ropes** or **compensation chains** that hang from the bottom of the car to the bottom of the counterweight. These compensate for the changing weight of the hoist ropes as the car moves through the hoistway. In our fleet, compensation systems are found in buildings over roughly 100 feet of travel.

Common issues include: - Compensation rope/chain rubbing against hoistway components - Excessive sway or swinging, particularly in high-rise installations with long compensation runs - Worn or damaged compensation sheaves at the bottom of the hoistway

Report any compensation system irregularities to ThyssenKrupp.

Governor Rope Inspection

The **governor** is a speed-sensing safety device. If the car overspeeds, the governor mechanically trips the **safety gear** (wedge-shaped clamps on the car frame) that grips the guide rails and stops the car. The governor rope loops from the governor at the top of the hoistway to a tension sheave at the bottom, attached to the car.

Inspect the governor rope for the same conditions as hoist ropes: broken wires, corrosion, diameter reduction. Additionally, verify that the governor rope tension sheave at the pit level moves freely and that the governor rope is properly seated in the governor sheave groove. A governor rope that has jumped its groove will not activate the safety gear in an overspeed event.

Governor rope and safety gear testing is performed during periodic inspections by TSSA-qualified inspectors. See the Inspection Types and What They Check document for inspection schedules and requirements.

Brake System Issues

The traction elevator brake holds the car in position when the motor is not running and provides emergency stopping. In a traction system, the brake is normally applied (spring-engaged) and released electrically (solenoid or motor-operated) when the elevator is about to move. This fail-safe design means that any loss of power results in the brake engaging, not releasing.

All brake adjustment and repair is contractor work. Technicians should understand brake symptoms to provide accurate reports.

Symptom: Car Creeps or Drifts at Floor

After the car stops at a floor and the doors open, the car slowly drifts up or down. Passengers may notice the floor of the car is no longer level with the landing. This is a serious safety concern because it creates a trip hazard. In our fleet's incident data, out-of-level conditions at door openings are among the most frequently reported causes of passenger injuries.

Possible causes:

1. **Brake pad wear.** As the brake pads wear, the braking force decreases. The brake may hold on initial application but slowly release under the weight imbalance between the car and counterweight.
2. **Spring tension loss.** The springs that apply the brake may have weakened over time, reducing clamping force.
3. **Solenoid issues.** If the brake solenoid is not fully de-energizing when the car should be stopped, the brake may only partially apply. Check for voltage on the brake solenoid circuit when the car is stopped; there should be none.
4. **Contamination.** Oil or grease on the brake disc or drum reduces braking friction. This can happen from leaking seals (geared systems) or during maintenance if lubricant is applied carelessly.

Field action: If a car is drifting at floor level, take it out of service immediately. This is not a "monitor and report" situation. Document the drift direction and approximate speed, check for obvious contamination, and escalate to ThyssenKrupp urgently. Reference our incident data: trip-and-fall injuries from out-of-level conditions are a known pattern in our fleet. See Common Failure Modes by Elevator Type for statistics across device types.

Symptom: Hard Stops or Jerky Stops

Passengers report rough stops, noticeable jolts when the car reaches a floor, or a "grabbing" sensation during deceleration.

Possible causes:

1. **Brake engagement too early.** If the brake applies before the motor has fully decelerated the car, the mechanical stop is abrupt. This is a drive/brake timing issue.
2. **Brake adjustment too tight.** The brake gap (clearance between the brake pad and the drum or disc when released) may be set too small, causing the brake to drag during deceleration.
3. **Hydraulic brake damper failure.** Some traction brake systems use a small hydraulic damper to cushion brake application. If the damper fails, brake engagement becomes sudden.
4. **Drive deceleration profile.** The drive may be decelerating too aggressively, and the abruptness is not the brake itself but the speed curve. Check drive fault logs for clues.

Field action: Document the behavior and whether it occurs at all floors or specific floors. Escalate to ThyssenKrupp for brake and drive inspection.

Symptom: Brake Won't Release

The elevator is stopped and will not move. The brake is engaged and does not release when a call is registered.

Possible causes:

1. **Solenoid failure.** The electrical solenoid that pulls the brake open may have failed. Check for voltage at the solenoid; if voltage is present and the brake is not releasing, the solenoid is likely failed.
2. **Mechanical binding.** The brake mechanism may be physically stuck. Corrosion, foreign material, or a broken component can prevent the brake from opening.
3. **Control circuit fault.** The controller may not be sending the brake release signal. This could be a relay failure, a broken wire, or a safety circuit issue preventing the controller from commanding the brake open.

Field action: Verify voltage at the brake solenoid. Do not attempt to manually release the brake unless following the manufacturer's emergency manual release procedure (for passenger rescue situations only; see Emergency Response Protocols). Escalate to ThyssenKrupp.

Brake Gap Measurement

The **brake gap** is the clearance between the brake pad and the brake drum (or disc) when the brake is released. Too large a gap means the brake takes longer to engage (more drift). Too small a gap means the brake drags during operation (wasted energy, heat, premature wear). Brake gap measurement and adjustment is contractor work, but technicians should know what to look for:

- Listen for brake drag noise (a continuous rubbing or scraping sound during car travel)
- Note whether the car coasts past the landing before stopping (possible excessive gap)
- Report any burning smell from the brake area (overheating from dragging)

Controller and Electrical Problems

The **controller** is the brain of the elevator. It processes hall calls and car calls, sequences the door operation, communicates with the drive, monitors safety circuits, and logs faults. Controllers range from older relay-based panels (still found in some of our legacy installations) to modern microprocessor-based systems.

Fault Code Interpretation

Every microprocessor controller maintains a fault log. The specific codes vary by manufacturer (ThyssenKrupp, KONE, and Schindler each have their own coding systems), but general categories are common:

- **Safety circuit faults** (usually highest priority): a safety device has tripped. The fault code typically identifies which device.
- **Drive faults**: the VFD has reported an error. These include overcurrent, overvoltage, undervoltage, overtemperature, encoder loss, and ground fault.
- **Door faults**: a door failed to open or close within the expected time, or an interlock is not making.
- **Communication faults**: loss of communication between the controller and the drive, between the controller and hall fixtures, or between controllers in a group of elevators.
- **Position faults**: the controller has lost track of where the car is in the hoistway.

When recording fault codes, always note the **timestamp** and **frequency**. A single door fault in a week is normal. Twenty door faults per day on the same floor points to a specific mechanical issue with that landing's door equipment.

Relay-Based vs. Microprocessor Controllers

Our older installations still use **relay-based controllers** where logic is implemented through banks of electromechanical relays. Troubleshooting these requires reading the wiring diagram and tracing the relay logic step by step. Look for:

- Relays that are stuck (contacts welded closed) or not pulling in (coil open or no voltage)
- Burnt relay contacts (visible blackening or pitting)
- Loose wiring connections in the relay panel

Microprocessor controllers are more capable but require different skills. They provide diagnostic screens, fault logs, and sometimes remote monitoring. However, they are also more sensitive to power quality issues and electrical noise. When a microprocessor controller behaves erratically, check the power supply to the controller board before suspecting the board itself.

Common Electrical Faults

Intermittent shutdowns. The elevator randomly takes itself out of service, then returns to normal operation. This is one of the most frustrating symptoms because it is difficult to catch in the act. Common causes: - Loose wiring connections that make and break contact with vibration - Safety circuit devices that are marginally tripping (e.g., a door interlock that is almost, but not quite, making contact) - Power quality issues (momentary voltage sags or transients) - Overheating controller components (the elevator works fine in the morning and fails in the afternoon as the machine room warms up)

Random floor stops. The car stops at floors where no call was registered. Causes include: - Hall call button stuck or shorted - Wiring fault in the call registration circuit - Controller board fault interpreting phantom calls

Recall to lobby. The elevator returns to the designated floor (usually the lobby) and parks with doors open. This is often triggered by the fire service recall circuit. Check: - Smoke detectors in the elevator lobby and machine room (a dirty or malfunctioning detector can trigger recall) - The fire service key switch position (someone may have activated recall and not returned the switch to normal) - The Phase 1 recall relay on the controller

Power Quality Issues

Traction elevator drives are sensitive to power quality. Problems to check:

- **Voltage fluctuations.** Measure voltage on all three phases at the main disconnect. Voltage should be within plus or minus 10% of nominal. Large voltage swings cause drive faults and erratic operation.
- **Phase imbalance.** Measure voltage across all three phase pairs (A-B, B-C, A-C). A difference of more than 2% between the highest and lowest reading indicates phase imbalance, which causes motor overheating and drive faults.
- **Harmonic distortion.** VFDs both produce and are affected by harmonic distortion on the power supply. If multiple elevators or other large electrical equipment share the same power feed, harmonic interference can cause problems. This requires a power quality analysis by an electrical contractor.

Field action: Take voltage readings and record them. Report significant deviations. Power quality remediation is building owner/electrical contractor territory, but we need to identify it as the root cause so the right people are contacted.

Controller Replacement vs. Repair

When a controller is failing frequently and the installation is more than 20 years old, the question of replacement vs. repair comes up. This is a decision that involves Rocket Elevators management, the building owner, and our contractor. Factors to consider:

- Availability of replacement parts for the existing controller
- Frequency and cost of service calls
- Whether the controller meets current CSA B44 requirements
- Energy efficiency gains from a modern drive and controller
- Building plans (is a full modernization planned in the near future?)

Technicians should document patterns of recurring failures clearly, as this data supports the business case for modernization. Ontario Regulation 209/01 may also require certain upgrades when specific conditions are met.

Leveling and Stopping Accuracy

Leveling refers to the car's ability to stop precisely at the landing floor level. CSA B44 specifies the acceptable leveling tolerance. Poor leveling is not just a ride quality issue; it is a safety concern. Our fleet's incident data consistently shows that out-of-level conditions are one of the most common causes of reported

injuries. Passengers trip stepping up or down from a car that is not level with the landing, and material handling at freight elevators becomes hazardous when the car floor is not flush with the dock.

Symptom: Car Stops Above or Below Floor Level

The car consistently stops with a gap (up or down) between the car floor and the landing floor.

Possible causes:

1. **Encoder drift.** The position encoder may have developed an offset. If the encoder signal is not accurately reflecting the car's position, the controller commands the stop at the wrong location. This is the most common cause and is corrected by recalibrating the encoder or adjusting the leveling parameters in the controller.
2. **Brake timing.** If the brake applies slightly early or late relative to the drive stopping the motor, the car will overshoot or undershoot the floor. The interaction between drive deceleration and brake engagement is carefully tuned, and drift in either system affects leveling.
3. **Rope stretch.** Over time, hoist ropes stretch, changing the relationship between the car position and the encoder reading. Rope stretch is gradual and causes a slowly worsening leveling error. When ropes are replaced or re-tensioned, leveling parameters may need adjustment.
4. **Load-dependent leveling errors.** The car levels correctly when empty but overshoots or undershoots with a full load. This indicates the **load-weighing device** is not correctly compensating, or the drive is not adjusting its stopping profile based on load. Load-weighing calibration is contractor work.
5. **Floor selector or position sensor issues.** Some systems use physical vanes and sensors in the hoistway to determine position near landings. Bent vanes, misaligned sensors, or failed sensor components cause leveling errors at specific floors.

Adjustment Procedures

Leveling adjustments fall into two categories:

Software adjustments (controller parameters): Adjusting the floor positions stored in the controller, the deceleration profile, and the brake timing. These are done through the controller interface and require knowledge of the specific controller software. This is contractor work.

Physical adjustments (sensors and vanes): Adjusting the position of hoistway sensors or vanes that trigger deceleration and stopping. Our technicians can identify misaligned or damaged sensors and report them, but physical adjustments in the hoistway are contractor work due to the confined space and fall hazards.

Field action: When responding to a leveling complaint, measure the actual gap at the problem floor(s) using a straightedge or level. Record whether the car is above or below the landing, the approximate gap in millimeters, and whether the error is consistent or varies with load. Check whether the issue occurs at all floors or specific floors (specific floors point to physical sensor issues; all floors point to encoder, brake, or rope issues). Document and escalate to ThyssenKrupp with your measurements.

Door System Troubleshooting

Door problems are the most common source of service calls across all elevator types in our fleet. They account for more callbacks than motor, drive, brake, and rope issues combined. This is true industry-wide, not just in our fleet. The door system has many moving parts, operates thousands of cycles per day, and is

the component most exposed to passenger interaction (people holding doors, blocking doors, bumping doors with carts and luggage).

A traction elevator has two sets of doors at each landing: the **car doors** (attached to the elevator car and travel with it) and the **hall doors** (also called **hoistway doors**, fixed at each landing). The car door operator drives the car doors, which engage the hall doors through a **door clutch** mechanism. The hall doors have **interlocks** that must verify the doors are closed and locked before the elevator can move.

Symptom: Doors Won't Close

The doors remain open and the elevator does not respond to calls. This is the most frequent door complaint.

1. **Obstruction sensors.** The door has a sensor (infrared curtain, light beam, or mechanical edge) that detects obstructions. If the sensor is malfunctioning, it may falsely detect an obstruction and hold the doors open. Clean the sensor lenses. Check for misalignment. Verify that nothing is actually in the doorway (debris on the sill, a loose threshold plate).
2. **Door operator belt or chain.** The door operator uses a belt or chain to drive the doors along their track. A broken or slipped belt results in the motor running but the doors not moving. Listen for the door motor running without door movement.
3. **Track alignment.** The door panels ride in tracks at the top (hanger) and bottom (sill). If the tracks are bent, dirty, or obstructed, the doors may bind and fail to close. Inspect the sill track for debris (a common issue in construction buildings and freight applications). Check the hanger rollers for wear or damage.
4. **Door interlock.** The hall door interlock verifies the door is closed and locked. If the interlock mechanism is damaged or misaligned, the door may close physically but the interlock switch does not make contact, so the controller thinks the door is still open. See "Door Interlock Faults" below.
5. **Door operator adjustment.** The door operator has adjustable parameters for opening speed, closing speed, closing force, and hold-open time. If the closing force is set too low, the doors may stall against minor friction.

Field action: Clean door sill tracks, clean sensor lenses, remove obstructions, check hanger rollers and belt/chain tension. These are within our scope. If the door operator itself needs repair or replacement, or if the interlock mechanism needs adjustment, escalate to ThyssenKrupp.

Symptom: Doors Won't Open

The car arrives at a floor but the doors remain closed. Passengers are effectively trapped (although they can usually be rescued via the emergency intercom or phone).

1. **Door operator failure.** The motor or controller in the door operator may have failed. Listen for any sound from the operator when the car arrives at a floor.
2. **Door clutch.** The **clutch** (also called the **vane** or **pickup**) on the car door engages the hall door so both move together. If the clutch fails to engage the hall door roller, the car doors may open but the hall doors remain closed, or neither set opens. Check the clutch mechanism for damage or misalignment.
3. **Retiring cam.** Some installations use a **retiring cam** mechanism that unlocks the hall door as the car arrives. If the cam is worn or misaligned, the hall door remains locked even though the car is at the landing.
4. **Controller fault.** The controller may not be sending the door open command. Check the controller for door-related fault codes.

Field action: If passengers are trapped, follow the Emergency Response Protocols for entrapment procedures. For diagnosis, check the clutch engagement and retiring cam visually. Report findings and escalate as needed.

Symptom: Doors Close Too Hard or Too Fast

Passengers or building staff report that the doors close with excessive force or speed. CSA B44 specifies maximum closing force limits (measured in the kinetic energy of the door). Excessive closing force is both a code compliance issue and a safety issue; door strikes are a common cause of reported injuries in our fleet.

1. **Closing force setting.** The door operator has an adjustable closing force parameter. If it has been set too high (sometimes done to overcome binding doors), reduce it to the correct value. The closing force must not exceed the CSA B44 limit.
2. **Closing speed setting.** Similarly, the closing speed may be set too high. Adjust to the recommended speed for the door type and weight.
3. **Door check (mechanical damper).** Some door systems use a mechanical damper to cushion the doors as they approach the closed position. A failed damper allows the doors to slam shut.
4. **Nudging mode.** If the doors have been held open past their time limit, the system enters **nudging mode**, which closes the doors at reduced speed with an audible warning. If nudging is too aggressive, it may feel like the doors are closing too hard. Nudging parameters are adjustable.

Field action: Closing force and speed adjustments are within our scope for minor adjustments. Use a door force gauge to verify compliance with CSA B44 limits. If the door operator mechanism itself needs repair, escalate.

Symptom: Door Interlock Faults

The **door interlock** is a safety device that verifies every hoistway door is closed and locked before the elevator can move. The interlock circuit is part of the safety chain. A broken interlock circuit stops the elevator.

This is safety-critical. Never bypass a door interlock for any reason. If an interlock is preventing the elevator from operating, find out why. The interlock is doing its job.

Common interlock issues:

- **Interlock switch misalignment.** The switch contacts do not fully close when the door is closed. This can happen from door track settling, hanger wear, or impact damage.
- **Worn interlock contacts.** After thousands of cycles, the switch contacts wear. Resistance increases, and eventually the circuit opens.
- **Interlock wiring.** The interlock circuit connects all hall doors in series. A wiring fault at any floor breaks the entire circuit. If the elevator stops with an interlock fault, identify which floor's interlock is the problem by checking the interlock contacts at each landing (or using the controller diagnostics if available).

Field action: Identify which floor's interlock is causing the fault. Visually inspect the interlock mechanism for obvious damage or misalignment. Do not attempt to adjust interlock mechanisms; this is contractor work because incorrect interlock adjustment can create a condition where the elevator runs with an unlocked door.

Hall Door vs. Car Door Issues

When diagnosing door problems, determine whether the issue is with the car door or the hall door:

- If the problem occurs at **every floor**, it is almost certainly a car door issue (since the car door travels with the elevator and is the same at every landing).
- If the problem occurs at **one specific floor**, it is almost certainly a hall door issue at that landing.
- If the problem is **intermittent** across floors, it may be a clutch engagement issue (the car door clutch does not consistently engage the hall door rollers).

This distinction saves significant diagnostic time and helps the contractor arrive prepared with the right parts.

Ride Quality Issues

Ride quality complaints are subjective, but they often point to real mechanical issues. Take them seriously, especially when multiple passengers or building staff report the same symptom.

Symptom: Vibration During Travel

Vibration felt in the car during travel (distinct from machine room vibration, which is covered in Motor and Drive Issues):

1. **Guide shoe or roller wear.** The car rides on guide rails using either sliding **guide shoes** (with replaceable gibs) or **roller guides**. Worn guide shoes or flat-spotted rollers allow the car to vibrate against the rails. Check the guide shoes or rollers for wear. Gib replacement is contractor work, but identifying the worn component is valuable.
2. **Rail alignment.** The guide rails run the height of the hoistway. If a rail joint is misaligned, the car will bump or vibrate at that specific point in its travel. Note whether the vibration occurs at the same position in the hoistway (same height) consistently.
3. **Rope vibration.** Long hoist ropes can resonate at certain frequencies, particularly in tall buildings. This manifests as a low-frequency swaying or oscillation. Compensation ropes can also contribute. Rope vibration is addressed through dampers or rope tension adjustment, both contractor work.
4. **Counterweight roller guides.** The counterweight rides on guides similar to the car. Worn counterweight rollers can transmit vibration through the ropes to the car. Check counterweight guide condition from the pit or machine room.

Field action: Ride the car and note where in the travel the vibration occurs. Is it constant, or at specific floors/heights? Report the location, severity, and character of the vibration (low-frequency sway vs. high-frequency buzz vs. single bump).

Symptom: Noise During Travel

Passengers or building staff report unusual sounds during elevator operation:

- **Clicking or ticking at regular intervals:** Likely a roller guide with a flat spot or a rope strand making contact with a hoistway component at the rope lay interval.
- **Squealing or screeching:** May be a belt-driven door operator (if the noise occurs during door operation), a sheave bearing, or sliding guide shoes without adequate lubrication.

- **Thumping at a specific height:** Misaligned rail joint or a hoistway obstruction that the counterweight or car is contacting.
- **Low hum or drone:** Motor bearing noise, gearbox noise, or transformer hum from the drive.
- **Wind noise at high speeds:** Common in tall buildings where the hoistway acts as a chimney. Not a mechanical issue, but building pressurization or hoistway vent adjustment can help.

Field action: Ride the car and characterize the noise: what it sounds like, when it occurs (during acceleration, at constant speed, during deceleration, at specific floors), and how loud it is. This characterization helps the contractor arrive with the right diagnostic equipment.

Symptom: Jerky Acceleration or Deceleration

Passengers feel the car lurch when starting or stopping, rather than smooth acceleration and deceleration:

1. **Drive tuning.** The VFD has acceleration and deceleration profiles that control how smoothly the motor ramps up and down. If these parameters are incorrect (too aggressive or poorly tuned), the car jerks at speed transitions. This is contractor work requiring drive-specific expertise.
2. **Encoder issues.** If the speed feedback from the encoder is noisy or intermittent, the drive cannot smoothly control the motor. This creates jerky corrections as the drive hunts for the correct speed.
3. **Load-weighing calibration.** The drive adjusts its torque output based on the car load. If the load-weighing device is miscalibrated, the drive applies too much or too little torque for the actual load, causing jerky starts (particularly noticeable when the car is fully loaded vs. empty).

Field action: Note whether jerkiness occurs during starts, stops, or both. Note whether it is worse with heavy loads. Check for encoder connection integrity. Report findings for contractor drive tuning.

Mapping Ride Quality Complaints to Components

When a ride quality complaint comes in, use this mapping to focus your diagnosis:

COMPLAINT	PRIMARY SUSPECT	SECONDARY SUSPECT
Vibration, all floors	Guide shoes/rollers	Motor or sheave bearings
Vibration, one spot	Rail joint at that height	Hoistway obstruction
Car sways/oscillates	Rope resonance	Roller guide clearance
Jerky starts/stops	Drive tuning	Load weighing
Noise, doors	Door operator	Clutch engagement
Noise, travel	Roller guides or rail joints	Sheave or bearing
Rough ride, loaded	Load weighing	Brake drag

When to Escalate

Clear boundaries between what our field technicians handle and what requires a licensed contractor protect both our team and our building owners. Ontario Regulation 209/01 and CSA B44 define what work requires

certified elevator mechanics, and TSSA enforces these requirements. Getting this wrong creates liability for Rocket Elevators.

What Our Field Technicians Handle

Our 14 technicians across six Ontario zones are trained and authorized to perform:

- **Visual inspections:** Rope condition (observation, not adjustment), sheave groove wear, brake condition, guide shoe condition, machine room environment, hoistway cleanliness
- **Door adjustments (minor):** Cleaning sill tracks, cleaning obstruction sensors, adjusting door closing speed and force within CSA B44 limits, replacing door gibs
- **Cab equipment:** Replacing car call buttons, floor indicators, position displays, emergency phone testing, cab lighting, fan operation
- **Safety circuit diagnosis:** Identifying which safety device is tripped (but not resetting or adjusting safety devices)
- **First-line electrical checks:** Voltage measurements, contactor verification, fuse checks, loose connection tightening at terminal blocks
- **Documentation and reporting:** Fault code recording, measurement recording, incident documentation
- **Environmental checks:** Machine room temperature, ventilation, cleanliness, lighting, access

What Requires a Licensed Contractor

Escalate to ThyssenKrupp (or the appropriate contractor for the installation) for:

- **Any work on the drive or controller internals.** This includes parameter changes, firmware updates, board replacement, and drive tuning
- **Brake adjustment or repair.** Brake gap setting, pad replacement, spring tension adjustment, solenoid replacement
- **Any rope work.** Rope replacement, tension equalization, rope shortening, compensation system adjustment
- **Sheave work.** Re-grooving, replacement, bearing replacement
- **Motor work.** Bearing replacement, winding repair, coupling replacement, motor replacement
- **Safety device testing or adjustment.** Governor calibration, safety gear testing, buffer testing, interlock mechanism adjustment
- **Leveling adjustments.** Encoder recalibration, position sensor adjustment, controller parameter changes
- **Any work that requires the car to be on the car top or in the hoistway.** Our technicians work from the machine room and the landing; they do not ride car tops or enter the pit (except for visual inspection with proper lockout/tagout)
- **Modernization or alteration.** Any modification to the original installation, including controller replacement, drive upgrades, or door operator replacement

The Escalation Process

1. Complete your diagnosis and documentation. The more detail you provide, the faster the contractor resolves the issue.
2. Determine urgency:

3. **Emergency:** Entrapment, safety device failure, car running with open doors, freefall event. Call ThyssenKrupp's 24/7 emergency line immediately. Follow our Emergency Response Protocols.
4. **Urgent:** Car out of service in a building with no other elevator, leveling issue creating a trip hazard, door closing force exceeding CSA B44 limits. Contact ThyssenKrupp for same-day or next-day response.
5. **Standard:** Performance issues, noise complaints, intermittent faults, routine repairs. Submit a service request through our standard process. Include your full diagnostic report.
6. Notify the building contact about the expected timeline.
7. Update the device status in our system if the car is taken out of service.
8. Follow up within the timeframe specified for the urgency level.

Contractor Contact Information

- **ThyssenKrupp Elevator (traction and MRL systems):** Primary contractor for all traction elevator service. Available 24/7 for emergencies. Standard service requests through our account portal.
- **KONE (hydraulic systems and modernizations):** For mixed-use buildings where traction and hydraulic units coexist, coordinate with both contractors.
- **Schindler (specialty installations and observation elevators):** Some observation elevators in our fleet are Schindler installations and require Schindler-specific parts and expertise.

Cross-References

- **Common Failure Modes by Elevator Type** for patterns and failure rates across the fleet, including comparisons between geared, gearless, and MRL traction systems
- **Safety Code Quick Reference** for CSA B44 rope inspection criteria, leveling tolerances, and door force limits
- **Emergency Response Protocols** for entrapment procedures, equipment failure response, and emergency communication
- **Hydraulic Elevator Maintenance Procedures** for hydraulic-specific systems (not covered in this guide)
- **Inspection Types and What They Check** for TSSA inspection categories, schedules, and how to prepare for periodic inspections

This document is reviewed quarterly by the Field Operations Manager and updated as fleet composition changes, contractor procedures are revised, or incident patterns reveal new troubleshooting priorities. All technicians are required to review updates within two weeks of publication. Questions or corrections should be directed to the Field Operations Manager.